

External Institution: Harvard SEAS — Computational Robotics Group

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MASTER'S THESIS

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Certifiable Trajectory Optimization for Locomotion via Semidefinite Relaxation

Problem description:

Nonlinear discretized dynamics and nonconvex geometric constraints yield a highly nonconvex optimization landscape; in contact-rich regimes, the discontinuity of contact forces further degrades it [4]. Classical approaches typically rely either on nonlinear programming (which is highly sensitive to initialization) [5] or on mixed-integer formulations [1] (which incur worst-case exponential complexity).

Locomotion requires reasoning about nonlinear dynamics, discontinuous contacts, and nonconvex geometry (self/obstacle avoidance). These properties create hard optimization landscapes and brittleness to initialization. In contrast, Moment-SOS semidefinite relaxation offers: (1) a convex lower bound that certifies suboptimality; (2) principled extraction of candidate trajectories; and (3) natural ways to exploit problem sparsity (chain-like temporal structure and robot-specific sparsity from kinematic chains and separable contacts) [2, 6]. Specialized first-order SDP solvers (including GPU-based ADMM variants) make these relaxations practically viable [3]. This thesis translates these ideas into legged locomotion on real hardware, using a certifiable trajectory optimization framework for locomotion tasks via sparse semidefinite relaxation.

Tasks:

- Review relevant literature on Lie-group based dynamics modeling for legged locomotion, including contact modeling and polynomial optimization methods on manifolds.
- Derive and formulate discrete-time Lie-group variational dynamics and express objectives and constraints as polynomial polynomialized optimization problems exploiting system structure.
- Develop sparse Moment-SOS / semidefinite relaxations that leverage temporal, kinematic, and contact sparsity for efficient and certifiable trajectory optimization.
- Evaluate certifiable solvers with trajectory extraction and validate the approach in simulation with respect to performance, robustness, runtime, and certification gap.

Bibliography:

- [1] Robin Deits and Russ Tedrake. Footstep planning on uneven terrain with mixed-integer convex optimization. In 2014 IEEE-RAS International Conference on Humanoid Robots (Humanoids), 2014.
- [2] Shucheng Kang, Guorui Liu, and Heng Yang. Global contact-rich planning with sparsity-rich semidefinite relaxations. In Robotics: Science and Systems (RSS), 2025.
- [3] Shucheng Kang, Xiaoyang Xu, Jay Sarva, Ling Liang, and Heng Yang. Fast and certifiable trajectory optimization. In International Workshop on the Algorithmic Foundations of Robotics (WAFR), 2024.
- [4] Tad McGeer et al. Passive dynamic walking. Int. J. Robotics Res., 9(2):62–82, 1990.
- [5] Michael Posa, Cecilia Cantu, and Russ Tedrake. A direct method for trajectory optimization of rigid bodies through contact. The International Journal of Robotics Research, 33(1):69–81, 2014.
- [6] Sangli Teng, Ashkan Jasour, Ram Vasudevan, and Maani Ghaffari. Convex geometric motion planning on Lie groups via moment relaxation. In Robotics: Science and Systems (RSS), 2023.

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